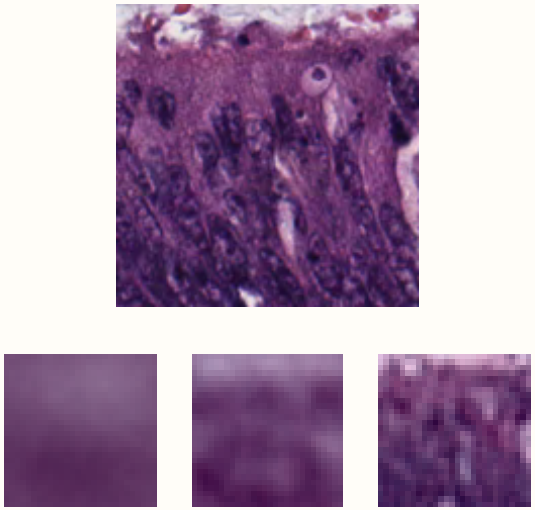
Learning PCA

How can a dataset choose its own coordinate system?

`fit · transform · inverse_transform`

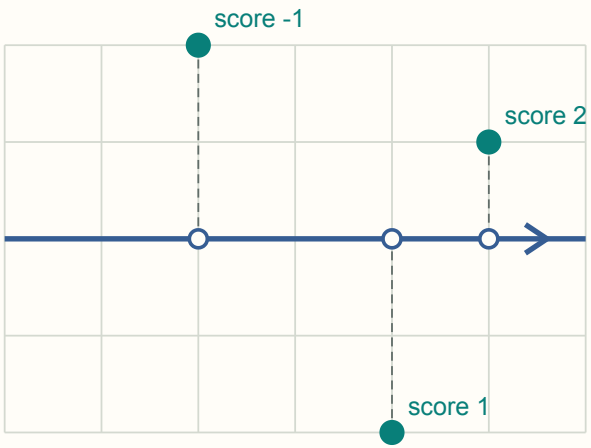


One validation image. Directions learned from training.

Kather et al. (2016) · CC BY 4.0

RECALL WEEK 7

A Shared Direction, Individual Scores



One direction shared by all observations

Component v

One direction shared by all observations.

Score z

One observation's coordinate along v.

Flip direction and scores

> v = [1, 0] · scores = [2, -1, 1]. Reconstructions stay [2, 0], [−1, 0], [1, 0].

Changing both signs preserves z v and the reconstruction.

FOUR CENTERED ROWS · TWO CORRELATED FEATURES

Correlated Features Reveal a Direction

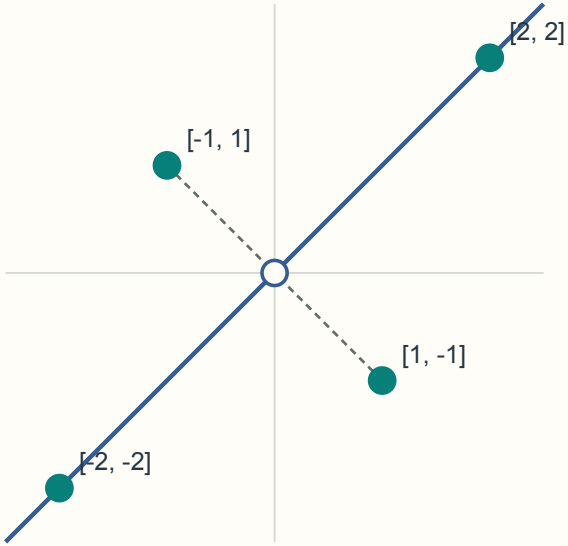
$$X = \begin{bmatrix} 2 & 2 \\ -2 & -2 \\ 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\mu = [0, 0]$$

$$C = \begin{bmatrix} 10/3 & 2 \\ 2 & 10/3 \end{bmatrix}$$

$X_c = X - \mu$. Here $X_c = X$.

Sample covariance divides $X_c^T X_c$ by $n - 1 = 3$.



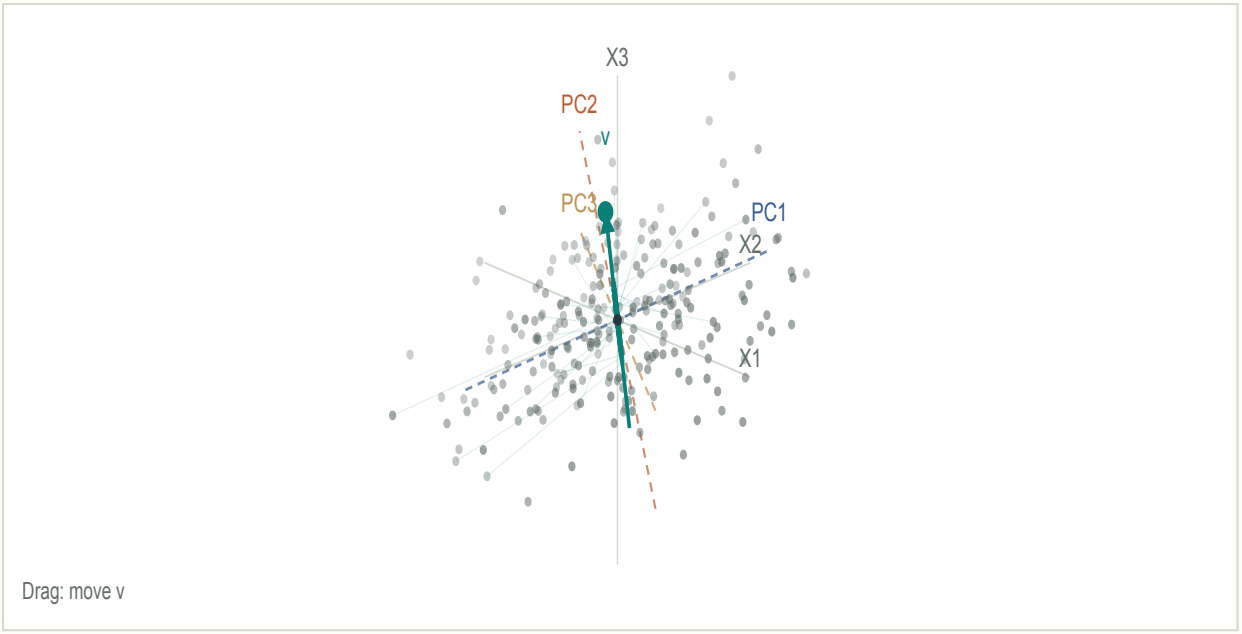
Keep the diagonal direction: $v_1 = [1, 1] / \sqrt{2}$

▶ Which direction should one component keep?

A principal direction can combine features instead of following a coordinate axis.

EXPLORE A UNIT DIRECTION IN THREE DIMENSIONS

A Direction Changes the Variance



240 centered samples · 3 features · direction arrows enlarged
v = [0.422, -0.527, 0.738] · ‖v‖ = 1.000

Maximum · PC1 | Minimum · PC3

Direction azimuth −51.3°

Direction elevation 47.5°

PC direction weights a_i^2

PC1 · λ_1 = 9.00 3.1%

PC2 · λ_2 = 3.00 0.1%

PC3 · λ_3 = 0.60 96.8%

Signed a: [0.176, 0.035, 0.984]

Projection variance 0.8622

λ_3 = 0.60 λ_1 = 9.00

Variance 0.8622; weights 3.1%, 0.1%, 96.8%.

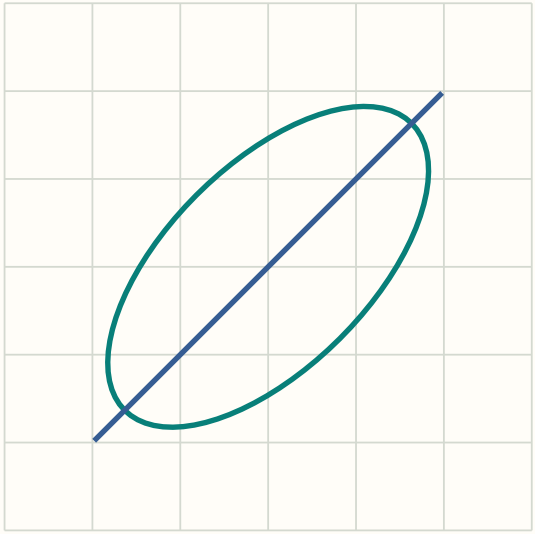
$\lambda_3 \le \mathrm{Var}(Xc\ v) \le \lambda_1$, with $\|v\| = 1$. The extremes align with PC3 and PC1.

NumPy · eigh

Columns are eigenvectors · ascending eigenvalues

CENTERED DATA → COVARIANCE → MAXIMUM SCORE VARIANCE

Maximum Variance Gives an Eigenvector



Unit direction: v = [cos θ, sin θ]
Same covariance as the four-row example.

$$C = X_c^T X_c / (n - 1)$$

$$\mathrm{Var}(X_c\, v) = v^T\, C\, v$$

Maximize $v^T C v$ with $v^T v = 1$

$$\Rightarrow C\, v = \lambda\, v$$

Unit direction angle 45°

Score variance = 5.333. Maximum 16/3 at 45°; perpendicular variance 4/3.

v is a direction; λ is the training-score sample variance along that direction.

NumPy · eigh

Columns are eigenvectors · ascending eigenvalues

THE NUMPY CALL STUDENTS IMPLEMENT

Call eigh, Then Select Component Rows

$C = Q\ \mathrm{diag}(\lambda)\ Q^T$

$Q^TQ = I$ · Q stores orthonormal directions in columns.

```
candidate_variance, axes = np.linalg.eigh(C)
```

Input C: (d, d), symmetric. Outputs: (d,) ascending variances and (d, d) direction columns.

Direction columns

Q

(d, d)

→

Sort and keep k

Q_k

(d, k)

→

Transpose to rows

$V = Q_k^T$

(k, d)

Variances [4/3, 16/3]

Descending indices [1, 0]

Keep each eigenvalue paired with its column; eigenvector signs may flip.

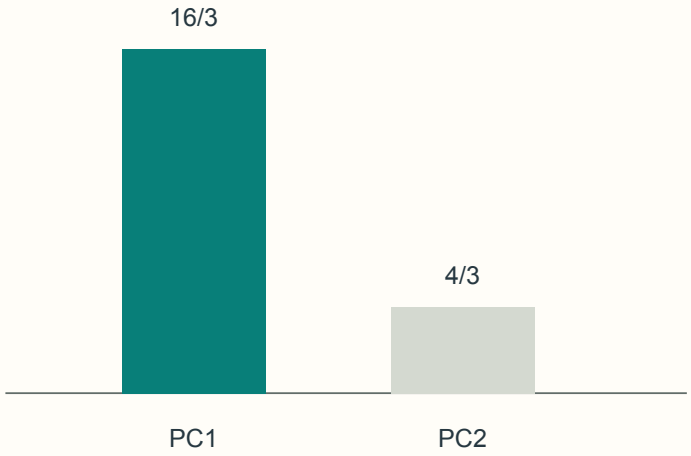
NumPy · eigh

Columns are eigenvectors · ascending eigenvalues

EIGENVALUES MEASURE THE SPREAD OF SCORES

Explained Variance Has a Fixed Denominator

Same example: eigenvalues are score variances



$$\mathrm{Var}(z_j) = \lambda_j$$

$$\mathrm{ratio}_j = \lambda_j \,/\, \mathrm{trace}(C)$$

Retained components 1

Keep 1: ratios = [0.8]. Captured = 80% of total 20/3.

Total = 16/3 + 4/3 = 20/3.

Use all training features even for a truncated solver.

For this example, the leading direction retains 80% of training variance.

PCA geometry reference

Instructor reference example

STATE LEARNED ONCE

A Fitted Object Keeps Its Coordinate System

`PCA(k).fit(X_train)`

`mean_`	(d,)
`components_`	(k, d)
`explained_variance_`	(k,)
`explained_variance_ratio_`	(k,)
`n_samples_`	training row count
`n_features_in_`	input width d

fit

Learns from training observations and returns self.

transform / inverse_transform

Reuse the saved mean and directions.

Learned attributes end in an underscore.

New observations enter the coordinate system defined by training.

IMPLEMENTATION CHECKPOINT 1 · COVARIANCE AND EIGH ARE YOUR WORK

Eight Expressions Complete Fit

01

Center training rows

`(n, d) → Xc, mean`

02

Build covariance

`covariance_matrix`
`(n, d) → (d, d)`

03

Call NumPy eigh

`(d, d) → values, axes`

04

Order variances

`largest first`

05

Select direction rows

`keep columns, then .T`

06

Clip selected variances

`tiny negatives → 0`

07

Compute total variance

`all squared centered entries / (n − 1)`

08

Compute ratios

`selected / total`

The teacher supplies validation, solver routing, image SVD and state assignment.

Your small-data fit explicitly builds covariance and extracts eigenvectors with NumPy.

IMPLEMENTATION CHECKPOINT 2

Two Methods Reuse Week 7

09 · transform(X)

Use your existing `project`.

$$Z = (X - \mu)V^T$$

`(m, d) → (m, k)`

10 · inverse_transform(Z)

Use your existing `reconstruct`.

$$\hat{X} = ZV + \mu$$

`(m, k) → (m, d)`

Both methods use the same saved mean and component rows.

ONE LEARNED ORIGIN

Validation Uses the Training Origin

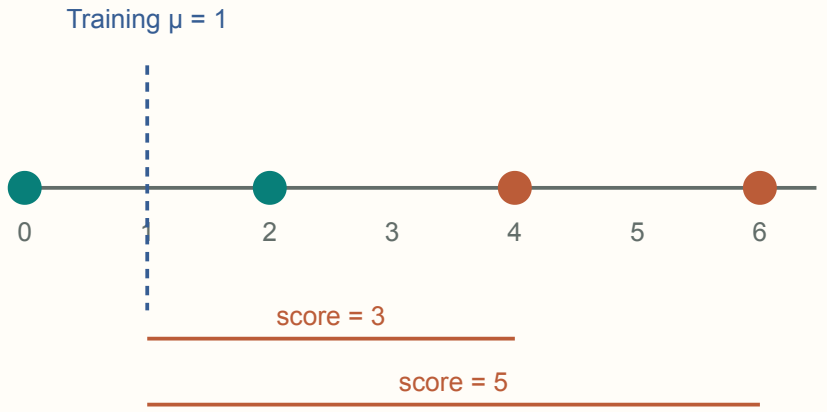
Training: [0, 2]

Saved mean = 1.

Validation: [4, 6]

Subtract the saved training mean.

```
Scores = [3, 5]
```



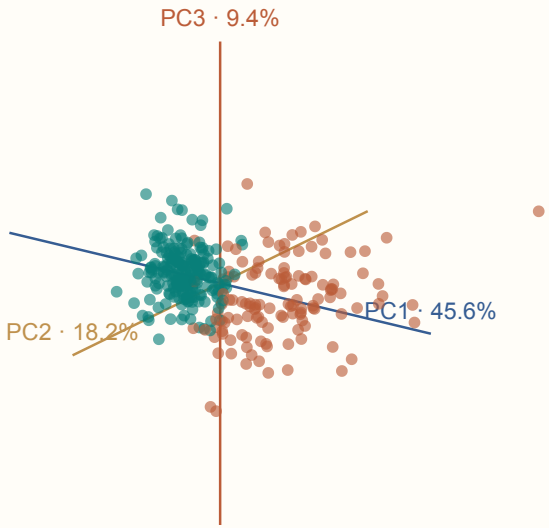
The validation batch does not move the origin.

▶ What if validation is centered again?

Changing the batch does not change the fitted model.

WDBC · 30 NAMED FEATURES

Thirty Features, Three Coordinates



Drag to rotate. Teal: benign. Rust: malignant.

Observations

Training · 341

Camera yaw 35°

Camera elevation 20°

Reset view

341 train observations. PC1–3 capture 73.2% of training variance. Same standardized training coordinates; class labels only color the points.

Three coordinates show one view of a 30-feature dataset.

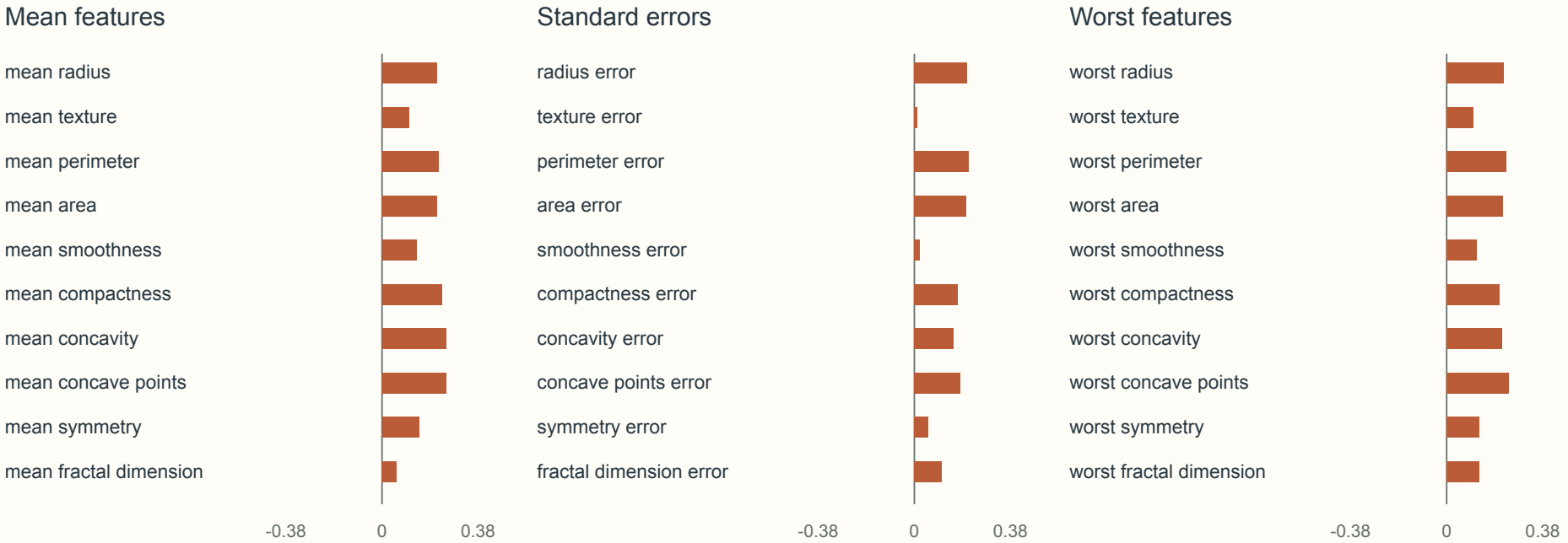
UCI · Breast Cancer Wisconsin (Diagnostic)

Reference calculation · Week 6 training split

SIGNED COEFFICIENTS IN STANDARDIZED FEATURE SPACE

A Component Combines Named Features

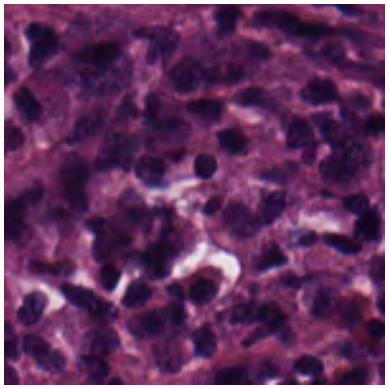
Component PC1 Flip all coefficient signs PC1 · 45.6% variance · unit-length direction · common scale across PCs



Read combinations of related features, with the chosen sign convention in mind.

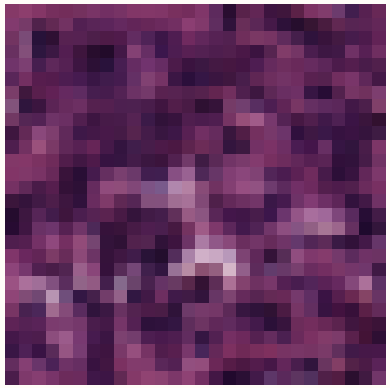
KATHER 2016 · EIGHT TISSUE CLASSES

The Same PCA API Fits Images



Original
150 × 150 RGB

→



Model input
28 × 28 RGB

→

2,352

features per image

`flatten, then /255`

3,000 training images	**300 shared components**	$\lambda_j = s_j^2 / (n - 1)$

The teacher's SVD helper supplies directions and variances. PCA receives no labels.

Fit once on training pixels, then reuse the same PCA object.

Kather et al. (2016) · CC BY 4.0

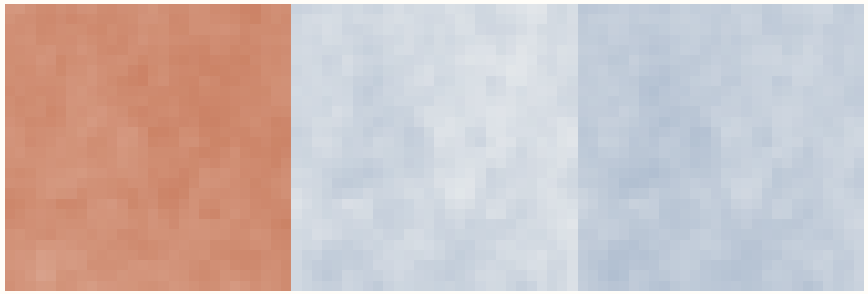
ONE LEARNED DIRECTION AT A TIME

Image Components Have Positive and Negative Weights

Component PC2

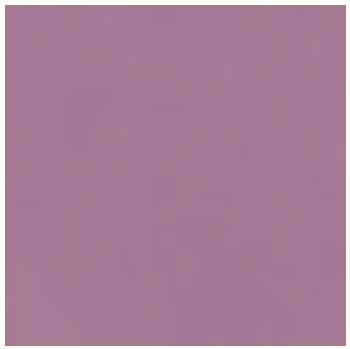
Score in training standard deviations

+2.00 σ

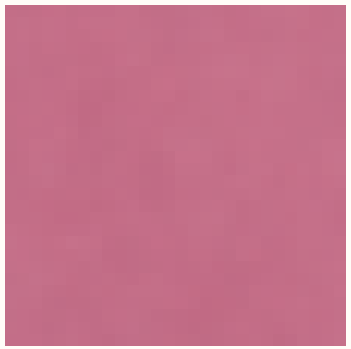


Signed coefficients: R · G · B

blue = negative · white = 0 · red = positive



Training mean μ



Mean + 2.00σ × PC2

$\sigma_j = \sqrt{\lambda_j}$. RGB images clip only for display. Signed maps share one scale.

A component describes a coordinated change across pixel channels.

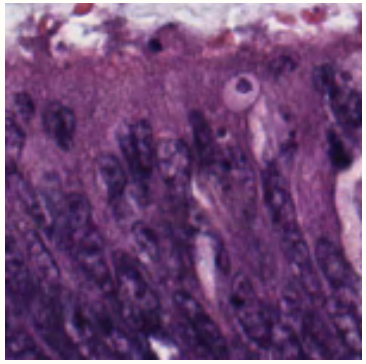
Kather et al. (2016) · CC BY 4.0

ONE TRAINING FIT · EIGHT FIXED VALIDATION EXAMPLES

A Learned Basis Reconstructs New Images

Validation tile: 01_TUMOR · #107

Components: 30



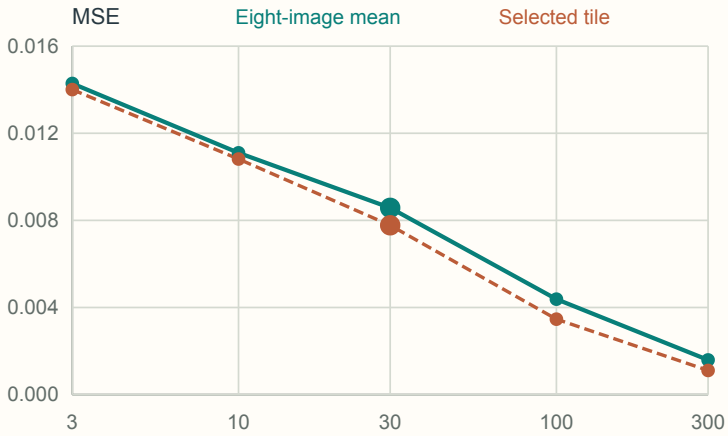
Original
150 × 150



Model input
28 × 28



30 PCs · 28 × 28



This tile's MSE: 0.00791. Eight-image mean: 0.00872. Error is measured before display clipping.

More directions preserve more of the measured input, with the same fitted basis.

Kather et al. (2016) · CC BY 4.0

Reference calculation · training basis · validation tiles

WORKING CODE AND EVIDENCE YOU CAN EXPLAIN

Verify, Interpret, and Connect to Week 9

One cumulative package

PCA plus all Week 1–7 functions and tests.

Reproducible results

State summary, 3D scores, named coefficients, components and reconstructions.

Open the student worksheet

- Which variance belongs in the denominator?
- Can a component and its scores both flip?
- What does a new batch change?
- Do three PCs prove good classification?

Week 9 learns class probabilities from these PCA scores.